

# Lecture 1: Basic Set Theory

Junnan Zhang

Paula and Gregory Chow Institute for Studies in Economics  
Xiamen University

Fall, 2026

Topics to be covered today

- ▶ Sets
- ▶ Functions
- ▶ Real Numbers

- 1 Sets
- 2 Functions
- 3 Real Numbers

- ▶ Set theory is the language of all mathematics
- ▶ In economics, sets are everywhere: consumption sets, production sets, sets of players, sets of equilibria, etc.

## Definition 1

- ▶ A set is, informally speaking, a collection of objects, which we call **elements**.
- ▶ If  $x$  is an element of the set  $S$ , we write  $x \in S$ . We also say  $x$  **belongs to**  $S$ .  
If  $x$  is not an element of  $S$ , we write  $x \notin S$ .

We use curly brackets when describing a set. We can list all its elements:

- ▶  $\{1, 2, 3\}$
- ▶  $\{1, 2, \{1, 3\}\}$

Or we can specify the properties that its elements satisfy:

- ▶  $\{x : x \text{ is irrational}\}$ : the set of all  $x$  such that  $x$  is irrational
- ▶  $\{n \in \mathbb{N} : n^2 < 7\}$ : the set of all natural numbers  $n$  such that  $n^2 < 7$

## Definition 2

- ▶ We say two sets  $A$  and  $B$  are **equal** if they have the same elements, and we write  $A = B$ .
- ▶ An **empty set** is a set with no elements and it is denoted by  $\emptyset$ .
- ▶ If  $\forall x \in A, x \in B$ , we say  $A$  is a **subset** of  $B$  and we write  $A \subset B$ .
- ▶ The set of all subsets of  $A$  is called the **power set** of  $A$  and is denoted by  $\mathcal{P}(A)$ .

We usually prove  $A = B$  by proving  $A \subset B$  and  $B \subset A$ .

- ▶  $\{\emptyset\}$  is not an empty set because  $\emptyset \in \{\emptyset\}$
- ▶  $\emptyset \subset A$  for all sets  $A$
- ▶ If  $A = \{1, 2\}$ , then  $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$
- ▶  $\emptyset \in \mathcal{P}(A)$  for all sets  $A$



## Proposition 3

*Set inclusion is transitive: if  $A \subset B$  and  $B \subset C$ , then  $A \subset C$ .*

*Proof:*

## Definition 4

- ▶ The **union** of  $A$  and  $B$  is defined by

$$A \cup B := \{x : x \in A \text{ or } x \in B\}$$

- ▶ The **intersection** of  $A$  and  $B$  is defined by

$$A \cap B := \{x : x \in A \text{ and } x \in B\}$$

- ▶ Two sets  $A$  and  $B$  are **disjoint** if  $A \cap B = \emptyset$

# Union and Intersection of Sets

## Venn Diagram

## Definition 5

- ▶ If  $\mathcal{F}$  is a family of sets, the union of all sets in  $\mathcal{F}$  is defined by

$$\bigcup \mathcal{F} := \{x : x \in A \text{ for at least one } A \in \mathcal{F}\}$$

- ▶ Similarly, the intersection of all sets in  $\mathcal{F}$  is defined by

$$\bigcap \mathcal{F} := \{x : x \in A \text{ for every } A \in \mathcal{F}\}$$

- ▶ If  $\mathcal{F} = \{A_\lambda : \lambda \in J\}$ , then

$$\bigcup \mathcal{F} = \bigcup_{\lambda \in J} A_\lambda \qquad \bigcap \mathcal{F} = \bigcap_{\lambda \in J} A_\lambda$$

## Proposition 6

Let  $A, B, C$  and  $B_\lambda$  (for  $\lambda \in J$ ) be sets. Then

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

$$A \cup (B \cup C) = (A \cup B) \cup C \quad A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \subset A \cup B \quad A \cap B \subset A$$

$$A \subset B \Leftrightarrow A \cup B = B \quad A \subset B \Leftrightarrow A \cap B = A$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap \bigcup_{\lambda \in J} B_\lambda = \bigcup_{\lambda \in J} (A \cap B_\lambda) \quad A \cup \bigcap_{\lambda \in J} B_\lambda = \bigcap_{\lambda \in J} (A \cup B_\lambda)$$

Prove some using the definitions

### Definition 7

- ▶ The **difference** of  $A$  and  $B$  is defined by

$$A \setminus B \text{ (or } A - B) := \{x : x \in A \text{ and } x \notin B\}$$

- ▶ If  $U$  is a set that contains all objects being considered in a certain context (a universal set), then  $U \setminus A$  is called the **complement** of  $A$  in  $U$ , and is denoted by  $A^c$

Examples:

- ▶ If  $U = \mathbb{R}$  and  $A = (0, 1)$ , then  $A^c = (-\infty, 0] \cup [1, +\infty)$
- ▶ If  $U = [0, 1]$  and  $A = (0, 1)$ , then  $A^c = \{0, 1\}$

### Proposition 8 (De Morgan's Laws)

Let  $A$ ,  $B$ , and  $A_\lambda$  (for  $\lambda \in J$ ) be sets. Then

$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

$$\left( \bigcup_{\lambda \in J} A_\lambda \right)^c = \bigcap_{\lambda \in J} A_\lambda^c \qquad \left( \bigcap_{\lambda \in J} A_\lambda \right)^c = \bigcup_{\lambda \in J} A_\lambda^c$$

Demonstrate using Venn diagrams

## Definition 9

- ▶ An **ordered pair** whose first member is  $x$  and second member is  $y$  is denoted by  $(x, y)$
- ▶ The basic property of ordered pairs is that  $(x, y) = (a, b)$  iff  $x = a$  and  $y = b$
- ▶ The Cartesian product of two sets  $A$  and  $B$  is defined by

$$A \times B := \{(a, b) : a \in A, b \in B\}$$

- ▶ We can also define  $n$ -tuples  $(a_1, a_2, \dots, a_n)$ . Then the Cartesian product of  $n$  sets is defined by

$$\prod_{i=1}^n A_i := \{(a_1, a_2, \dots, a_n) : a_i \in A_i, \forall i = 1, 2, \dots, n\}$$



# Cartesian Product

## Examples

►  $\{1, 2\} \times \{3, 4\} = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$

►  $\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}}_{n \text{ times}}$

- ▶ Natural numbers:  $\mathbb{N} = \{1, 2, \dots\}$
- ▶ Integers:  $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$
- ▶ Rational numbers:  $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z}, q \neq 0\}$
- ▶ Real numbers:  $\mathbb{R}$
- ▶ Complex numbers:  $\mathbb{C}$

- 1 Sets
- 2 Functions
- 3 Real Numbers

# What is a Function?

## Definition 10

Let  $X$  and  $Y$  be two sets. A **function** or **map** or **mapping**  $f$  from  $X$  to  $Y$  associates to every element  $x \in X$  an element  $y \in Y$ , denoted by  $f(x)$ . We write

$$f: X \rightarrow Y$$

$$x \mapsto f(x)$$

We call  $X$  the **domain** of  $f$  and  $Y$  the **codomain** of  $f$ .

We call  $x$  an **argument** of  $f(x)$ , and  $f(x)$  the **image** of  $f$  at  $x$ .

### Definition 11

- ▶ Let  $S \subset X$ . The **image** of  $S$  under  $f$  is defined by  $f(S) = \{f(x) : x \in S\}$ .
- ▶ The image of  $X$  is called the **range** of  $f$ .
- ▶ Let  $T \subset Y$ . The **inverse image** of  $T$  under  $f$  is defined by  $f^{-1}(T) := \{x : f(x) \in T\}$ .
- ▶ We say  $f$  is **one-one** or **injective** if for every  $y \in Y$ , there is at most one  $x \in X$  such that  $y = f(x)$ .
- ▶ We say  $f$  is **onto** or **surjective** if for every  $y \in Y$ , there exists  $x \in X$  such that  $y = f(x)$ .
- ▶ If  $f$  is both one-one and onto, we say  $f$  is **bijective**.

# Image and Inverse Image

## Examples

## Proposition 12

$$f(C \cup D) = f(C) \cup f(D) \qquad f\left(\bigcup_{\lambda \in J} A_\lambda\right) = \bigcup_{\lambda \in J} f(A_\lambda)$$

$$f(C \cap D) \subset f(C) \cap f(D) \qquad f\left(\bigcap_{\lambda \in J} A_\lambda\right) \subset \bigcap_{\lambda \in J} f(A_\lambda)$$

$$f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D) \qquad f^{-1}\left(\bigcup_{\lambda \in J} A_\lambda\right) = \bigcup_{\lambda \in J} f^{-1}(A_\lambda)$$

$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D) \qquad f^{-1}\left(\bigcap_{\lambda \in J} A_\lambda\right) = \bigcap_{\lambda \in J} f^{-1}(A_\lambda)$$

$$(f^{-1}(A))^c = f^{-1}(A^c) \qquad \text{no general results for } f$$

$$A \subset f^{-1}(f(A))$$

## Definition 13

- ▶ If  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$ , then the **composition** function  $g \circ f : X \rightarrow Z$  is defined by

$$(g \circ f)(x) = g(f(x))$$

- ▶ The **identity** function on  $X$  is defined by  $Id_X(x) = x$  for all  $x \in X$ .
- ▶ We say  $f : X \rightarrow Y$  is **invertible** if there exists  $g : Y \rightarrow X$  such that  $f \circ g = Id_Y$  and  $g \circ f = Id_X$ . In this case,  $g$  is the **inverse** of  $f$  and is denoted by  $f^{-1}$ .

- ▶ Example:  $f(x) = x^2$ ,  $g(x) = \sin(x)$ . Then  $(g \circ f)(x) = \sin(x^2)$  and  $(f \circ g)(x) = \sin^2(x)$ .
- ▶ The composition operator is associative:  $h \circ (g \circ f) = (h \circ g) \circ f$ .



## Proposition 14

*$f$  is bijective iff  $f$  is invertible.*

*Proof:* Suppose that  $f$  is bijective. Then, for every  $y \in Y$ , there is a unique  $x \in X$  such that  $f(x) = y$ . This allows us to define  $g : Y \rightarrow X$  by  $f(g(y)) = y$ . Therefore,  $f \circ g = Id_Y$ . Now pick any  $x \in X$  and let  $y = f(x)$ . The definition of  $g$  implies that  $g(y) = x$ . Therefore,  $g(f(x)) = x$  and thus  $g \circ f = Id_X$ .

Suppose  $f$  is invertible. Then for any  $y \in Y$  there is an  $x \in X$  given by  $x = f^{-1}(y)$  such that  $f(x) = f(f^{-1}(y)) = Id_Y(y) = y$ . Now suppose there is another  $x' \in X$  such that  $f(x') = y$ . Then  $f^{-1}(f(x')) = Id_X(x') = x'$ . Since  $f^{-1}(f(x')) = f^{-1}(y) = x$ ,  $x = x'$ . Therefore,  $f$  is bijective.

## Definition 15

- ▶ Two sets  $X$  and  $Y$  are **equivalent** if there exists a bijection  $f : X \rightarrow Y$ . We write  $X \sim Y$ .
- ▶ With every set  $A$  we associate a **cardinal number** or **cardinality** to indicate the “number” of elements in  $A$ . Two sets have the same cardinality if they are equivalent.
- ▶  $Y$  has higher cardinality than  $X$  if there exists an injection from  $X$  into  $Y$ .

## Definition 16

- ▶ A set is finite if it is empty or it is equivalent to the set  $\{1, 2, \dots, n\}$  for some  $n \in \mathbb{N}$ . In this case, it has cardinality  $n$ . A set is **infinite** if it is not finite.
- ▶ A set is **denumerable** (or countably infinite) if it is equivalent to  $\mathbb{N}$ . A set is **countable** if it is denumerable or finite. If a set is denumerable, we say it has cardinality  $d$ .
- ▶ A set is **uncountable** if it is not countable. If a set is equivalent to  $\mathbb{R}$ , we say it has cardinality  $c$ .

## Proposition 17

- ① *Any subset of a countable set is countable.*
- ② *The set of rational numbers  $\mathbb{Q}$  is denumerable.*
- ③ *The set of even numbers is equivalent to  $\mathbb{N}$ . The set  $(0, 1)$  is equivalent to  $\mathbb{R}$ .*
- ④ *The set  $\mathbb{N}$  and  $(0, 1)$  are not equivalent.*
- ⑤ *The product of two countable sets is countable.*
- ⑥ *The cardinality of  $\mathbb{R}^n$  is  $c$ .*
- ⑦ *The product of two sets of cardinality  $c$  has cardinality  $c$ .*
- ⑧ *The union of a countable family of countable sets is countable.*

- 1 Sets
- 2 Functions
- 3 Real Numbers

# The Real Number System

- ▶ Most functions we encounter in economics are real-valued functions
- ▶ It is important to know some basic properties of the real numbers
- ▶ It is also key to the study of convergence later on

# The Real Number System

The **Real Number System** is a set of objects called **Real Numbers** together with two binary operations called **addition** (+) and **multiplication** ( $\times$ ), a binary relation called **less than** ( $<$ ), and two elements called **zero** (0) and **unity** (1), that satisfies a set of axioms:

► properties of addition

A1  $a + b = b + a$

A2  $(a + b) + c = a + (b + c)$

A3  $a + 0 = 0 + a = a$

A4 there is exactly one real number, denoted by  $-a$  such that  $a + (-a) = (-a) + a = 0$

- properties of multiplication

- A5  $a \times b = b \times a$

- A6  $(a \times b) \times c = a \times (b \times c)$

- A7  $a \times 1 = 1 \times a = a$ , and  $1 \neq 0$

- A8 if  $a \neq 0$  then there is exactly one real number, denoted by  $a^{-1}$  such that  $a \times a^{-1} = a^{-1} \times a = 1$

- the distributive property

- A9  $a(b + c) = ab + ac$

- Axioms A1 - A9 are called algebraic axioms



## Definition 18

- ▶ Define **subtraction** by  $a - b = a + (-b)$
- ▶ Define **division** by  $a/b = a \times b^{-1}$  for  $b \neq 0$

## Theorem 19

- 1 If  $a + c = b + c$ , then  $a = b$ .
- 2 If  $ac = bc$  and  $c \neq 0$ , then  $a = b$ .

*Proof:* Since  $a + c = b + c$ ,  $(a + c) + (-c) = (b + c) + (-c)$ . It follows from A2 that  $a + (c + (-c)) = b + (c + (-c))$ . By A4,  $a + 0 = b + 0$ . By A3,  $a = b$ . The second claim can be proved in a similar way.

## Theorem 20

*If  $c \neq 0$  and  $d \neq 0$ , then*

- ①  $a0 = 0, -(-a) = a$
- ②  $(c^{-1})^{-1} = c, (-1)a = -a$
- ③  $a(-b) = -(ab) = (-a)b$
- ④  $(-a) + (-b) = -(a + b)$
- ⑤  $(-a)(-b) = ab$
- ⑥  $(a/c)(b/d) = (ab)/(cd)$
- ⑦  $(a/c) + (b/d) = (ad + bc)/(cd)$

Prove some using Axioms A1-A9

- ▶ The Order Axioms: there is a subset  $P$  of the real numbers, called the set of **positive numbers** such that

A10 For any real number  $a$ , exactly one of the following holds:

$$a = 0 \quad \text{or} \quad a \in P \quad \text{or} \quad -a \in P$$

A11 If  $a \in P$  and  $b \in P$ , then  $a + b \in P$  and  $ab \in P$

- ▶ The “Less Than” Relation: define  $a < b$  to mean  $b - a \in P$ ,  $a \leq b$  to mean  $b - a \in P$  or  $b - a = 0$ ,  $a > b$  to mean  $a - b \in P$ , and  $a \geq b$  to mean  $a - b \in P$  or  $a = b$ .
- ▶  $a < b$  iff  $b > a$  and  $a \leq b$  iff  $b \geq a$

## Theorem 21

- ①  $a < b$  and  $b < c$  implies  $a < c$
- ② exactly one of  $a < b$ ,  $a = b$ , and  $a > b$  is true
- ③  $a < b$  implies  $a + c < b + c$
- ④  $a < b$  and  $c > 0$  implies  $ac < bc$
- ⑤  $a < b$  and  $c < 0$  implies  $ac > bc$
- ⑥  $0 < 1$  and  $-1 < 0$
- ⑦  $a > 0$  implies  $1/a > 0$
- ⑧  $0 < a < b$  implies  $0 < 1/b < 1/a$

*Similar results hold for  $\leq$*

Prove some using Axioms A1-A11

## Definition 22

If  $S$  is a set of real numbers, then

- 1  $a$  is an **upper bound** for  $S$  if  $x \leq a$  for all  $x \in S$ ;
- 2  $b$  is the **least upper bound** (or **l.u.b.** or **supremum** or **sup**) for  $S$  if  $b$  is an upper bound and  $b \leq a$  whenever  $a$  is an upper bound.
- 3 We call  $b$  a **maximum** of  $S$  if  $b$  is the l.u.b. for  $S$  and  $b \in S$ .

We can similarly define **lower bound**, **greatest lower bound** (or **g.l.b.** or **infimum** or **inf**), and **minimum**.

A set is **bounded above** if it has an upper bound and is **bounded below** if it has a lower bound.

## A12 Least Upper Bound Completeness Axiom

Suppose  $S$  is a nonempty set of real numbers that is bounded above. Then  $S$  has a l.u.b in  $\mathbb{R}$ .

## Corollary 23

*Suppose  $S$  is a nonempty set of real numbers that is bounded below. Then  $S$  has a g.l.b in  $\mathbb{R}$ .*

## Proposition 24

*Suppose  $S$  is a nonempty set of real numbers. Then  $b$  is the supremum of  $S$  iff*

- ①  $x \leq b$  for all  $x \in S$ , and
- ② for each  $\epsilon > 0$ , there exists  $x \in S$  such that  $x > b - \epsilon$ .

*Proof:*

## Theorem 25

*The set  $\mathbb{N}$  is not bounded above.*

*Proof:* Suppose  $\mathbb{N}$  is bounded above. By the Completeness Axiom, there is a supremum  $b \in \mathbb{R}$  such that  $n \leq b$  for all  $n \in \mathbb{N}$ . Since  $n + 1 \in \mathbb{N}$ ,  $n + 1 \leq b$  and thus  $n \leq b - 1$  for all  $n \in \mathbb{N}$ . This is a contradiction since  $b$  is no longer the least upper bound of  $\mathbb{N}$ .

## Theorem 26

- ① *For any  $x, y \in \mathbb{R}$ , if  $x < y$  then there exists a rational (irrational) number  $r$  such that  $x < r < y$ .*
- ② *For any  $x \in \mathbb{R}$  and  $\epsilon > 0$ , there exists a rational (irrational) number  $r$  such that  $0 < |r - x| < \epsilon$ .*



# The Archimedean Property

## Exercise

Prove that the supremum of  $\{x \in \mathbb{R} : x < 3\}$  is 3.

*Proof:*

### Definition 27

Let  $f : X \rightarrow Y$  be a **real-valued** function, that is,  $Y \subset \mathbb{R}$ .

Define upper bounds, the supremum, the maximum of  $f$  as the upper bounds, supremum, and maximum of its range  $f(X) \subset \mathbb{R}$ . The arguments  $x$  such that  $f(x)$  is the maximum are called the **maximizers** of  $f$ .

We say  $f$  is **bounded** if its range is bounded.

## Definition 28

If  $X \subset \mathbb{R}$ , then we say

- ▶  $f$  is **increasing** if  $x \leq x' \Rightarrow f(x) \leq f(x')$
- ▶  $f$  is **decreasing** if  $x \leq x' \Rightarrow f(x) \geq f(x')$
- ▶  $f$  is **monotonic** if it is increasing or decreasing
- ▶  $f$  is **strictly increasing** if  $x < x' \Rightarrow f(x) < f(x')$
- ▶  $f$  is **strictly decreasing** if  $x < x' \Rightarrow f(x) > f(x')$